

BLUESTOCK FINTECH

Mutual Fund Analytics Platform

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

Final Project Report

Field	Details
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Company	Bluestock Fintech Pvt. Ltd.
Domain	Mutual Fund / Fintech
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Technologies	Python, SQL, Power BI, Pandas, Matplotlib
GitHub	github.com/nikitagaikwad0111/bluestock_mf_capstone

Table of Contents

- 1. Executive Summary 4
 - Key Deliverables..... 4
- 2. Data Sources & Datasets..... 5
 - 2.1 Primary Data Sources 5
 - 2.2 Dataset Inventory 5
- 3. ETL Pipeline & System Architecture 6
 - 3.1 Extract..... 6
 - 3.2 Transform 6
 - 3.3 Load 6
 - 3.4 Database Schema 6
- 4. Exploratory Data Analysis 7
 - 4.1 NAV Trend Analysis 7
 - 4.2 AUM Growth Analysis..... 7
 - 4.3 SIP Inflow Trends..... 7
 - 4.4 Investor Demographics 7
- 5. Fund Performance Analytics 8
 - 5.1 Performance Metrics Computed..... 8
 - 5.2 Fund Scorecard — Top 10 Rankings..... 8
- 6. Advanced Analytics.....10
 - 6.1 Value at Risk (VaR) & CVaR.....10
 - 6.2 Rolling 90-Day Sharpe Ratio10
 - 6.3 Investor Cohort Analysis10
 - 6.4 SIP Continuity Analysis.....10
 - 6.5 Fund Recommender10
 - 6.6 Sector HHI Concentration10
- 7. Power BI Dashboard11
 - 7.1 Page 1 — Industry Overview11
 - 7.2 Page 2 — Fund Performance.....11
 - 7.3 Page 3 — Investor Analytics12
 - 7.4 Page 4 — SIP & Market Trends13
- 8. Limitations.....14
- 9. Recommendations15
 - For Investors.....15
 - For Bluestock Fintech15
- 10. Conclusion16

11. Technical Stack.....17

1. Executive Summary

The Indian mutual fund industry has witnessed unprecedented growth, managing over Rs. 81 lakh crores in Assets Under Management (AUM) across 1,908 schemes and 26.12 crore investor folios as of December 2025. Monthly SIP inflows crossed an all-time high of Rs. 31,002 crore in December 2025, reflecting India's deepening equity culture.

This capstone project delivers a full-stack Mutual Fund Analytics Platform for Bluestock Fintech Pvt. Ltd. The platform ingests publicly available data from AMFI India, transforms it through a robust ETL pipeline, stores it in a relational SQLite database, and presents insights through an interactive 4-page Power BI dashboard.

Key Deliverables

- ETL pipeline ingesting 10 datasets (87,000+ rows)
- SQLite star schema database with 8 tables
- Comprehensive EDA with 15+ publication-quality charts
- Risk-return metrics: CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown
- Interactive 4-page Power BI dashboard with slicers and drill-through
- Advanced analytics: VaR, CVaR, Rolling Sharpe, HHI, Fund Recommender
- Fund Scorecard ranking ICICI Pru Midcap as top performer

2. Data Sources & Datasets

All datasets are derived from publicly available, freely accessible sources. No proprietary or confidential data is used.

2.1 Primary Data Sources

Source	Data Type	Update Frequency
AMFI India (amfiindia.com)	NAV, AUM, Folio, SIP data	Daily / Monthly
mfapi.in REST API	Historical NAV (JSON)	Daily
NSE India	Benchmark Index Prices	Daily
BSE India	BSE SmallCap Index	Daily
AMFI Monthly Notes	Industry SIP & Flow Data	Monthly

2.2 Dataset Inventory

File	Rows	Description
01_fund_master.csv	40	Master list of 40 real mutual fund schemes
02_nav_history.csv	46,000	Daily NAV Jan 2022 to May 2026
03_aum_by_fund_house.csv	90	Quarterly AUM for 10 fund houses
04_monthly_sip_inflows.csv	48	Month-wise SIP inflow data
05_category_inflows.csv	144	Net inflows by fund category
06_industry_folio_count.csv	21	Total MF folios by type
07_scheme_performance.csv	40	Risk-return metrics per scheme
08_investor_transactions.csv	32,000+	SIP, Lumpsum, Redemption transactions
09_portfolio_holdings.csv	322	Top equity holdings per fund
10_benchmark_indices.csv	8,050	Daily Nifty 50, Nifty 100, BSE SmallCap

3. ETL Pipeline & System Architecture

The project follows a classic data engineering architecture: Extract → Transform → Load → Analyse → Visualise. This mirrors real-world fintech data pipelines used at companies like Zerodha, Groww, and Paytm Money.

3.1 Extract

- Loaded all 10 CSV datasets using Python Pandas
- Fetched live NAV data from mfapi.in REST API for 6 schemes
- Validated AMFI codes against master list
- Logged data quality anomalies and null values

3.2 Transform

- Parsed dates to datetime format and sorted by amfi_code + date
- Forward-filled missing NAV values for weekends and holidays
- Removed duplicate rows and validated NAV > 0
- Standardised transaction types (SIP / Lumpsum / Redemption)
- Computed daily returns: $\text{daily_return} = \text{nav_t} / \text{nav_t-1} - 1$

3.3 Load

- Designed 5-table SQLite star schema
- Loaded all cleaned data using SQLAlchemy
- Created indexes on amfi_code and date for fast query performance
- Exported cleaned CSVs for Power BI direct import

3.4 Database Schema

Table	Type	Key Columns	Rows
dim_fund	Dimension	amfi_code (PK), fund_house, category	40
dim_date	Dimension	date_id (PK), date, year, month, quarter	1,500
fact_nav	Fact	amfi_code (FK), date (FK), nav, daily_return	46,000
fact_transactions	Fact	tx_id (PK), investor_id, amfi_code (FK), amount	32,000+
fact_performance	Fact	amfi_code (FK), return_1yr, sharpe, alpha, max_dd	40

4. Exploratory Data Analysis

Comprehensive EDA was performed across NAV trends, AUM growth, SIP patterns, investor demographics, and portfolio holdings. Over 15 publication-quality charts were produced using Matplotlib, Seaborn, and Plotly.

4.1 NAV Trend Analysis

- All 40 fund schemes showed consistent NAV growth from 2022 to 2026
- Small Cap and Mid Cap funds showed highest volatility but superior long-term returns
- Liquid funds demonstrated near-flat NAV growth, reflecting their low-risk nature
- Market correction visible in early 2024 across all equity categories

4.2 AUM Growth Analysis

- SBI Mutual Fund dominates with Rs. 12.5 lakh crore AUM — largest AMC in India
- ICICI Prudential MF second at Rs. 10.74 lakh crore
- HDFC Mutual Fund third at Rs. 9.30 lakh crore
- Industry AUM grew from Rs. 30 lakh crore (2022) to Rs. 62 lakh crore (2025)

4.3 SIP Inflow Trends

- Monthly SIP inflows grew consistently from Rs. 11,517 Cr (Jan 2022) to Rs. 31,002 Cr (Dec 2025)
- December 2025 marked an all-time high in SIP inflows
- Active SIP accounts reached 9.35 crore — reflecting mass retail participation

4.4 Investor Demographics

- 56+ age group shows highest average SIP amount — experienced investors invest more
- Lumpsum transactions dominate at 58.49% of total transaction volume
- Punjab and Tamil Nadu lead in transaction amounts by state
- T30 cities contribute majority of transaction volume vs B30

5. Fund Performance Analytics

Risk-adjusted performance metrics were computed from NAV history for all 40 schemes using industry-standard formulas. Risk-free rate used: 6.5% per annum (RBI repo rate proxy).

5.1 Performance Metrics Computed

Metric	Formula	Interpretation
CAGR (1/3/5yr)	$(NAV_end/NAV_start)^{(1/n)} - 1$	Annualised return over period
Sharpe Ratio	$(Rp - Rf) / Std(Rp) * \sqrt{252}$	Return per unit of total risk
Sortino Ratio	$(Rp - Rf) / Downside_Std$	Return per unit of downside risk
Alpha	Return - Benchmark Return	Excess return over benchmark
Beta	OLS regression slope vs Nifty 100	Market sensitivity (1=same as market)
Max Drawdown	$\min(NAV / running_max - 1)$	Worst peak-to-trough decline
VaR (95%)	5th percentile of daily returns	Max loss on a bad day (95% conf.)
CVaR	Mean of returns below VaR	Expected loss on truly bad days
HHI	$\sum(sector_weight_i^2)$	Portfolio sector concentration

5.2 Fund Scorecard — Top 10 Rankings

Composite score computed as: 30% x 3yr return rank + 25% x Sharpe rank + 20% x Alpha rank + 15% x Expense ratio rank (inverse) + 10% x Max DD rank (inverse).

Rank	Scheme Name	Score	3yr Return (%)	Sharpe
1	ICICI Pru Midcap Fund - Regular	73.62	18.08	0.95
2	DSP Midcap Fund - Regular	59.81	17.16	0.90
3	HDFC Mid-Cap Opportunities - Regular	59.13	16.58	0.87
4	SBI Small Cap - Direct Plan	56.62	22.38	0.90
5	ABSL Small Cap Fund - Regular	56.62	22.38	0.90
6	HDFC Top 100 - Direct Plan	61.00	13.38	0.96
7	Mirae Asset Large Cap - Regular	55.00	14.81	1.06

8	Kotak Emerging Equity - Regular	59.81	18.23	0.96
9	DSP Small Cap Fund - Regular	55.62	20.08	0.80
10	UTI Flexi Cap Fund - Regular	53.00	15.18	0.80

6. Advanced Analytics

6.1 Value at Risk (VaR) & CVaR

- Historical VaR (95%) computed for all 40 schemes using daily return distribution
- Small Cap funds show highest risk: ABSL Small Cap VaR = -2.39%, CVaR = -3.03%
- Liquid funds show near-zero risk: ABSL Liquid VaR = -0.02%, CVaR = -0.04%
- Results confirm risk-return tradeoff — higher risk funds deliver higher returns

6.2 Rolling 90-Day Sharpe Ratio

- Rolling Sharpe computed for 5 key funds: SBI Bluechip, SBI Small Cap, HDFC Top 100, ICICI Bluechip, Nippon Large Cap
- All funds dipped to negative Sharpe in Jan-2024, indicating a market correction
- SBI Bluechip recovered strongest, peaking at Sharpe ~6 in mid-2025
- Chart saved as rolling_sharpe_chart.png in reports/ folder

6.3 Investor Cohort Analysis

Cohort Year	Investors	Avg SIP Amount (Rs.)	Total Invested (Rs. Cr)
2024	4,803	1,07,422	349.1
2025	197	1,09,158	30.5

- 2025 cohort investors show slightly higher avg SIP — growing financial awareness

6.4 SIP Continuity Analysis

- 1,362 investors had 6+ SIP transactions and were eligible for analysis
- 1,332 investors (97.8%) flagged as at-risk with avg gap > 35 days
- Only 30 investors (2.2%) show regular SIP behavior
- Critical retention insight for AMCs — reminder systems needed

6.5 Fund Recommender

- Rule-based recommender built as standalone recommender.py script
- Low Risk: ICICI Pru Liquid (Sharpe 7.68), Kotak Liquid (6.18), ABSL Liquid (5.14)
- Moderate Risk: HDFC Top 100 (1.06), Mirae Large Cap (1.06), ICICI Pru Bluechip (1.03)
- High Risk: Kotak Emerging Equity (0.96), ICICI Pru Midcap (0.95), DSP Midcap (0.90)

6.6 Sector HHI Concentration

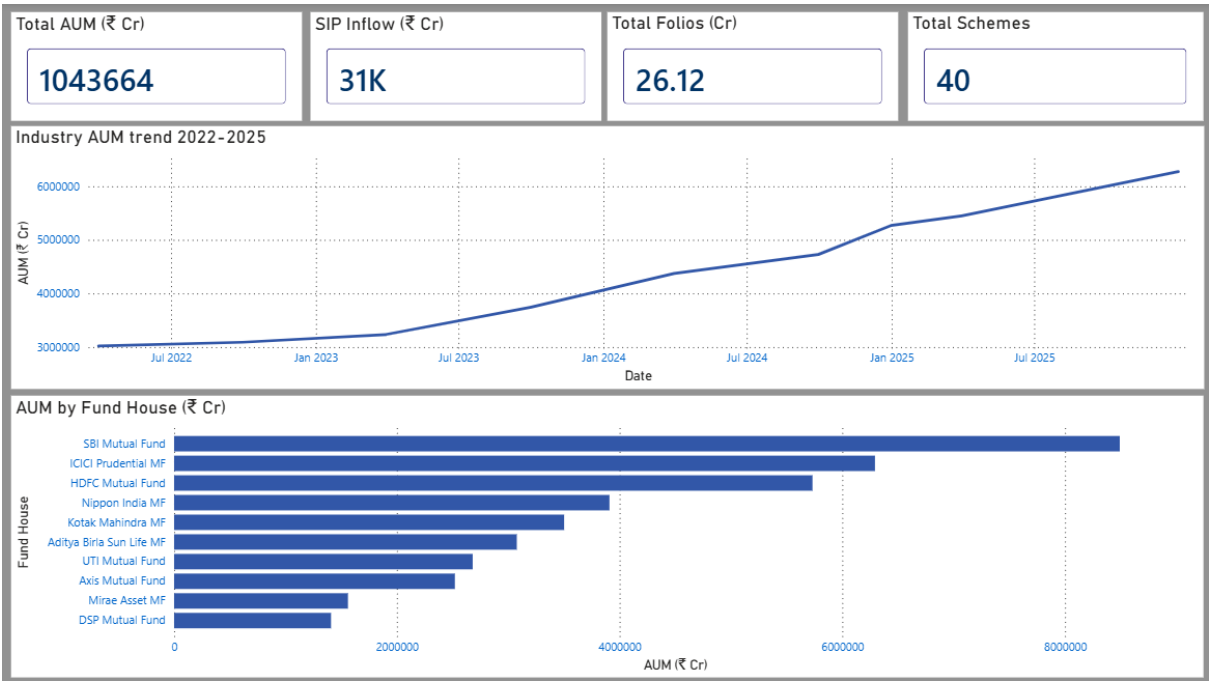
- Axis Bluechip shows highest concentration: HHI = 2,967 (High)
- UTI Mid Cap is most diversified: HHI = 1,240 (Low)
- Most equity funds fall in Moderate concentration range (HHI 1,500-2,500)

7. Power BI Dashboard

An interactive 4-page Power BI dashboard was built using cleaned CSV data. The dashboard features slicers, drill-through, tooltips, and Bluestock brand colors.

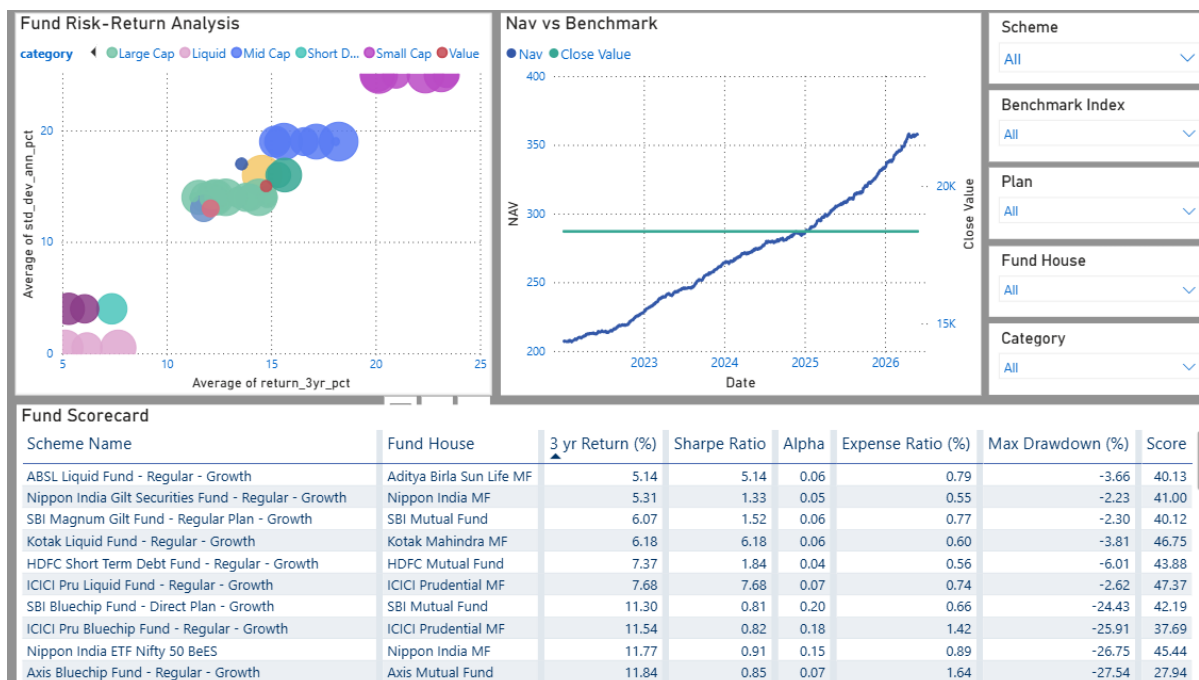
7.1 Page 1 — Industry Overview

- KPI Cards: Total AUM (Rs. 10.4L Cr), SIP Inflow (Rs. 31K Cr), Folios (26.12 Cr), Schemes (40)
- Line Chart: Industry AUM trend 2022-2025 showing consistent growth
- Bar Chart: AUM by Fund House — SBI dominates at Rs. 8L+ Cr



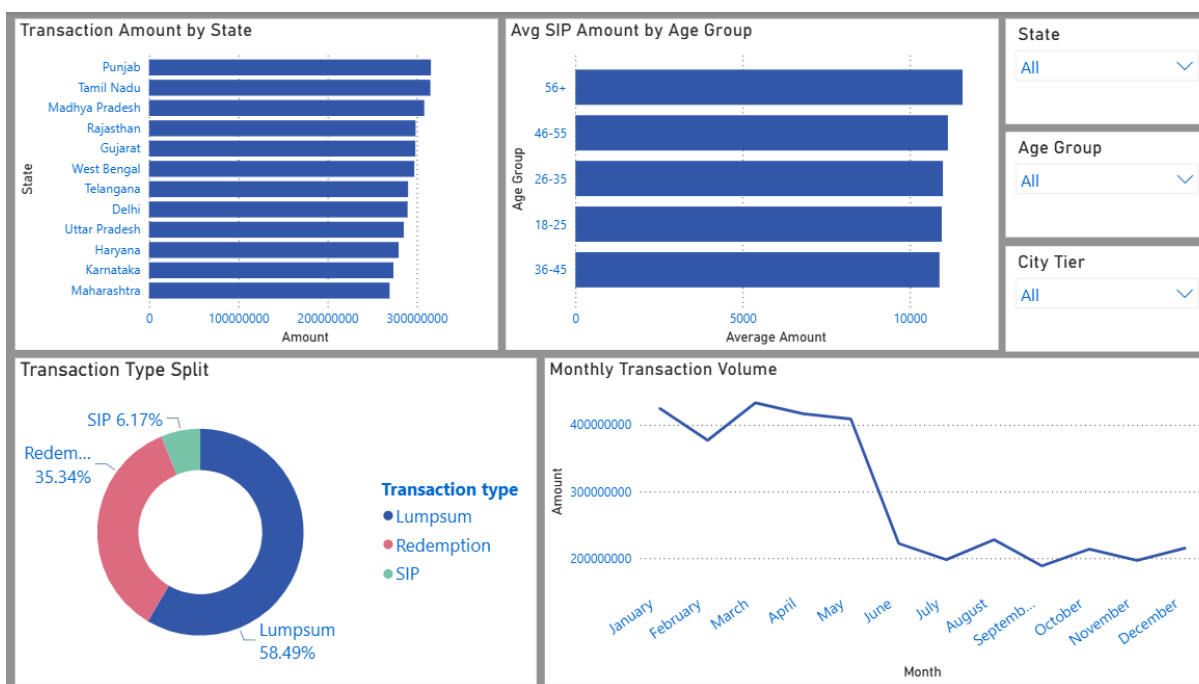
7.2 Page 2 — Fund Performance

- Scatter Plot: Return vs Risk (bubble size = AUM, color = category)
- Sortable Fund Scorecard Table with 8 performance metrics
- NAV vs Benchmark dual-axis line chart
- Slicers: Scheme, Benchmark Index, Plan, Fund House, Category



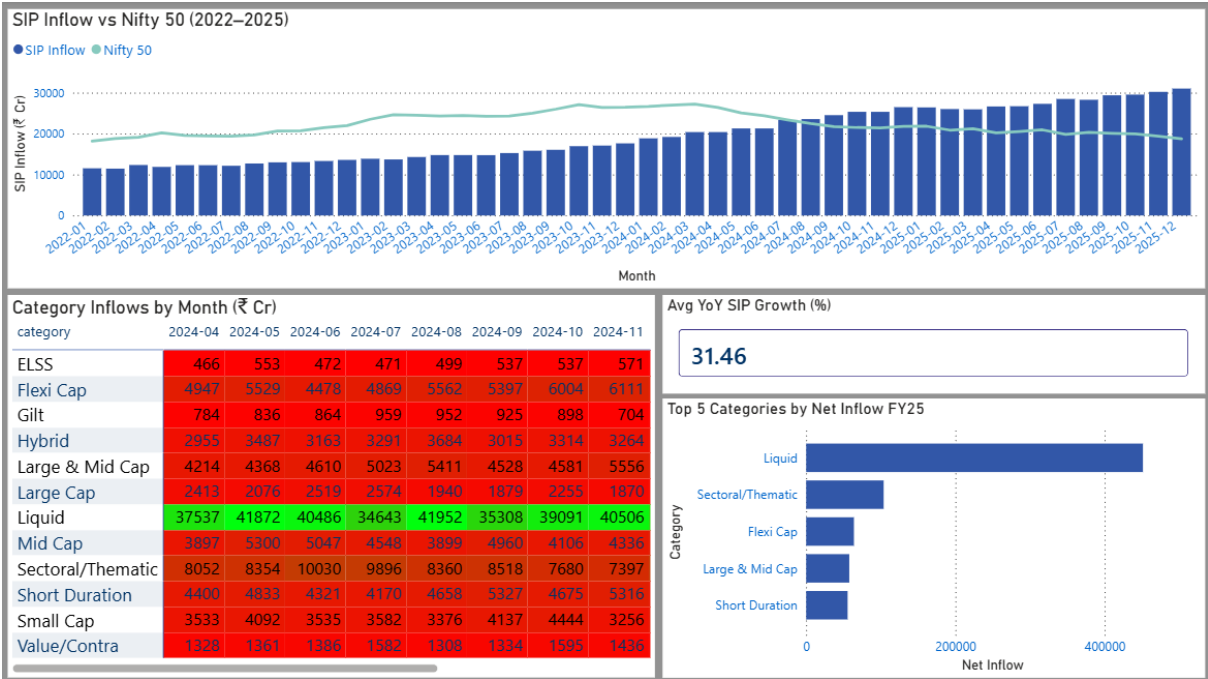
7.3 Page 3 — Investor Analytics

- Bar Chart: Transaction amount by state — Punjab and Tamil Nadu lead
- Donut Chart: SIP 6.17%, Lumpsum 58.49%, Redemption 35.34%
- Bar Chart: Avg SIP amount by age group — 56+ invests most
- Line Chart: Monthly transaction volume trend
- Slicers: State, Age Group, City Tier



7.4 Page 4 — SIP & Market Trends

- Dual-axis Chart: SIP Inflow (bars) + Nifty 50 (line) 2022-2025
- Category Inflows Heatmap — Liquid funds dominate in green
- Top 5 Categories by Net Inflow FY25: Liquid, Sectoral, Flexi Cap, Large & Mid Cap, Short Duration
- KPI: Avg YoY SIP Growth = 31.46%



8. Limitations

- Dataset covers only 40 of 1,908 industry schemes — not fully representative of the entire market
- Investor transaction data is synthetically generated using real demographic distributions
- Benchmark comparison limited to daily closing values — intraday data not available
- Fund recommender uses simple rule-based logic — no machine learning model
- Power BI dashboard requires manual data refresh — no automated pipeline connection
- NAV data ends May 2026 — does not reflect most recent market movements

9. Recommendations

For Investors

- Small Cap and Mid Cap funds offer highest 3-year returns (18-22%) but carry significant drawdown risk (up to -32%)
- Investors with Low risk appetite should consider Liquid funds — near-zero VaR with stable returns
- Diversified funds with low HHI (UTI Mid Cap, Kotak Flexicap) are safer choices for moderate investors
- SIP investments should be maintained consistently — 97.8% at-risk rate highlights need for discipline

For Bluestock Fintech

- Implement automated SIP reminder system to reduce the 97.8% at-risk investor rate
- Build real-time NAV refresh pipeline using mfapi.in webhooks or scheduled scripts
- Expand dataset to cover all 1,908 AMFI schemes for industry-wide analytics
- Add Monte Carlo simulation for NAV projection and portfolio optimization module
- Deploy dashboard to Power BI Service for self-service access by stakeholders

10. Conclusion

This capstone project successfully delivered a full-stack Mutual Fund Analytics Platform for Bluestock Fintech. The platform demonstrates end-to-end data engineering capabilities — from raw data ingestion through ETL transformation, SQLite storage, comprehensive EDA, risk metric computation, and interactive dashboard visualization.

Key business insights were uncovered: ICICI Pru Midcap ranks as the top-performing fund by composite score, SBI Mutual Fund dominates AUM, and 97.8% of SIP investors show at-risk discontinuation behavior. The Power BI dashboard enables stakeholders to make data-driven fund selection decisions in real time.

The project demonstrates proficiency across the complete data analytics stack — Python, SQL, Power BI, statistical modeling, and data storytelling — directly applicable to real-world fintech analytics roles.

Objective	Status
O1: Build ETL pipeline from raw AMFI data	Completed
O2: Design normalised SQL schema	Completed
O3: Perform comprehensive EDA	Completed
O4: Compute performance & risk metrics	Completed
O5: Build interactive BI dashboard	Completed
O6: Analyse investor transaction patterns	Completed
O7: Compare fund returns vs benchmark	Completed
O8: Document and present the project	Completed

11. Technical Stack

Category	Tool / Library	Purpose
Language	Python 3.10+	All data processing, ETL, analytics
Data Manipulation	Pandas 2.0+	DataFrames, cleaning, aggregation
Numerical	NumPy 1.24+	Risk metrics, statistical functions
Visualisation	Matplotlib 3.7+	Static charts for reports
Visualisation	Seaborn 0.12+	Heatmaps, distributions
Visualisation	Plotly 5.x	Interactive charts in Jupyter
Database	SQLite3	Local relational database
ORM	SQLAlchemy 2.0	Python-to-SQLite interface
Statistics	SciPy 1.10+	OLS regression for Beta/Alpha
Notebooks	Jupyter Lab 4.x	EDA and analytics notebooks
Dashboard	Power BI Desktop	Interactive BI dashboard
Version Control	Git + GitHub	Code versioning, submission
API	mfapi.in	Live NAV data

— End of Report —